A G O R A

A Sandbox for the Economics of Abundance

Owning the Machine

A Governance of Abundance for Europe

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How artificial intelligence threatens to split growth from livelihoods — and why Europe should respond by distributing ownership of the machine, not just income from it. A data-grounded proposal for Universal Basic Capital, stress-tested in a stock-flow-consistent model of the European economy.

A sandbox, not an oracle — please read this first

Every quantitative result in this document comes from AGORA, a transparent simulation that compares policies inside one consistent, accounting-complete model of the economy. These are scenario comparisons under explicit, inspectable assumptions — not forecasts. The value is in the direction and size of the trade-offs between policies, not in any single number. Every figure is reproducible; every assumption is sourced and can be changed. The model is calibrated to live Eurostat, AMECO, BIS and OECD data for 2019 and validated against 2010–2019 history (it reproduces realised GDP to ~1%).

Prepared with the AGORA engine · Germany & France detail, 26 EU economies calibrated · June 2026

Executive summary

Artificial intelligence is poised to do something no previous technology has done at this speed: raise output while reducing the share of that output paid to human labour. If the gains flow to whoever owns the AI capital — the models, the compute, the data — then Europe can grow richer and more unequal at the same time. This is the Great Decoupling. The usual answer, a cash transfer such as Universal Basic Income, treats the symptom (low income) but not the cause (concentrated ownership). You cannot offset a compounding stock with a flat flow.

This study argues — and shows, inside a fully accounting-consistent model — that the durable answer is predistribution of ownership: give every citizen a share of the AI capital stock through a citizens' sovereign wealth fund that pays a per-capita dividend. We call it Universal Basic Capital (UBC). The same engine that diagnoses the problem is used to test the cure, head-to-head against cash transfers and against doing nothing.

What the engine shows (Germany, 30-year AI-shift scenario, capital levy τ = 40%)

- The decoupling is real and measurable. With no policy, labour's share of output falls from 65.5% toward 30%, output still rises (to ~228 on a 2019=100 index), and at-risk-of-poverty climbs to ~21%.
- Cash helps income but never ownership. Cash UBI lowers the income-Gini (to 0.24) yet leaves the top-10% wealth share parked at its real level (~56%). UBC collapses it to ~20% — because it distributes the asset, not just the income. This wealth result holds regardless of our most uncertain assumption.
- The citizens' dividend compounds. A flat cash transfer stays flat; the fund's dividend overtakes it around year 10 and reaches roughly €81,545 per citizen by year 30 (vs €32,731 for cash), as citizens come to own ~79% of the capital stock.
- It need not cost growth. If the fund reinvests, the UBC economy ends about 1.11× the size of the cash economy — predistribution does not have to choke investment; it relocates who does the investing.
- Pooling halves the European gap. A purely national AI dividend widens the gap between richer and poorer member states; a pooled EU dividend cuts between-country inequality roughly in half.
- No single 'best' — but UBC owns the frontier. Across the growth–equality trade-off, UBC policies dominate both cash UBI and inaction. 'Do nothing' is off the frontier entirely.

> **The choice in one sentence**
> Europe can rent the AI economy from its owners, or it can own it. The evidence here says ownership is the only instrument that touches the dimension — wealth — where the AI transition concentrates power.

PART I

The problem: growth without shared prosperity

For two centuries, productivity growth and wages rose together: better tools made workers more valuable. Artificial intelligence breaks that link in a specific way. When a task is automated, the income it generated stops flowing to a worker and starts flowing to the owner of the system that replaced them. If automation outpaces the creation of new human tasks, labour's share of national income falls — and because capital ownership is far more concentrated than wages, the gains pool at the top even as the economy grows.

AGORA models this as an AI-shift scenario: a sustained capital-investment boom (grounded in observed frontier-compute growth of ~4.2× per year) together with a falling labour share. Run against Germany with no policy response, the model reproduces the decoupling at the household level. Output keeps climbing while labour's share collapses — the scissors in Figure 1 — and relative poverty rises accordingly.

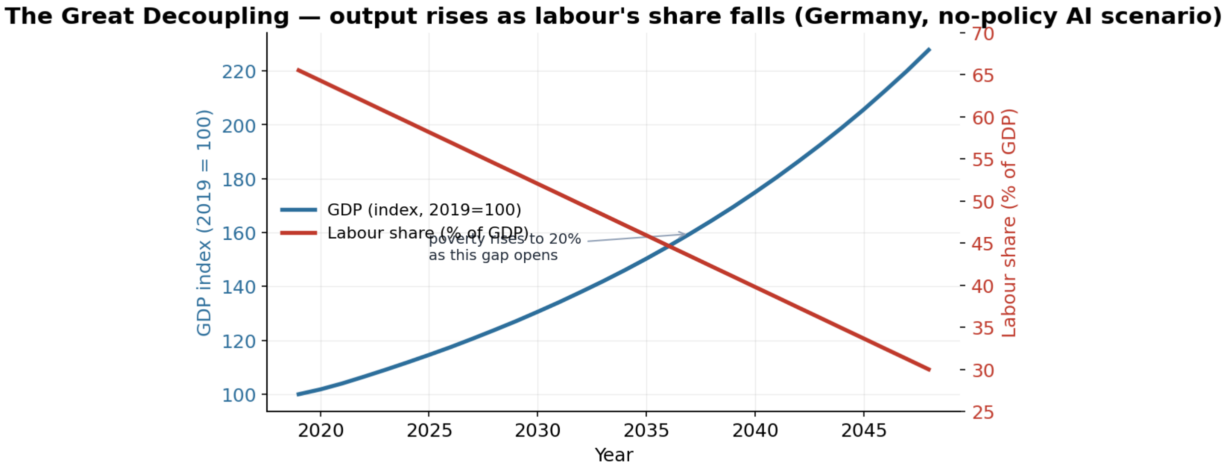

Figure 1 — Output and labour income pull apart. As AI raises productivity, GDP keeps rising while labour's share of it falls toward 30%.
Source: AGORA engine, AI-shift no-policy scenario, Germany, calibrated to Eurostat/AMECO 2019. Illustrative scenario, not a forecast.

Crucially, the shock is not a uniform drift. Using the real Eurostat input-output table for Germany (FIGARO), AGORA distributes the labour-share fall across sectors by their exposure to automation. The result (Figure 2) is stark: the labour share in ICT, finance and business services falls from 45% to ~2%, while construction — hard to automate — barely moves (63% to 51%). The aggregate decoupling is the sum of a few sectors automating hard. That matters for policy: blunt, economy-wide instruments are responding to a concentrated shock.

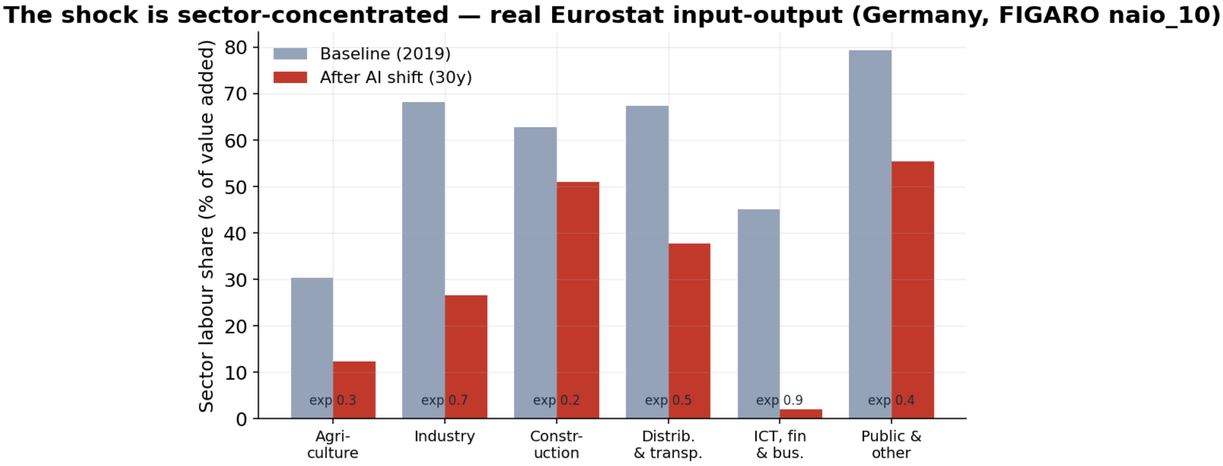

Figure 2 — A concentrated shock. High-exposure sectors shed labour share fastest; low-exposure sectors barely move.

Source: AGORA input-output module on real Eurostat FIGARO (naio_10_cp1700), Germany. Sectoral value added reconciles to GDP (consistency-gated).

> **Why income transfers alone cannot fix this — the flow-versus-stock problem**
>
> Cash UBI redistributes a flow (this year's income). But the AI advantage is a compounding stock: capital, network effects, and the data flywheel that grows as it is used. Hand out a flat flow and ownership — and therefore the compounding — stays exactly where it was. Owners keep pulling away; recipients get a fixed drip that rent and asset prices erode. To distribute a compounding advantage you must distribute the asset, not just the income it throws off.

There is also a fiscal warning. As the wage base shrinks, the tax revenue that funds public services and transfers erodes with it — precisely when demand for support rises. A system that taxes labour to fund redistribution is taxing the very thing AI is shrinking. This is a structural signal of the transition, independent of the inequality result.

PART II

Why the standard toolkit falls short

Faced with the decoupling, the most-discussed response is cash Universal Basic Income: an unconditional payment to every citizen. It is simple, dignified, and supported by trial evidence. In AGORA it does real good — it lowers the income-Gini and cuts poverty. But it has a structural blind spot. Look at wealth, not income (Figure 3). Cash UBI redistributes the income that capital throws off; it never changes who owns the capital. The top-10% wealth share stays at its observed level (~56% for Germany, from OECD data), while under UBC it falls to ~20%. Income compression fades if the transfer stops; ownership is permanent.

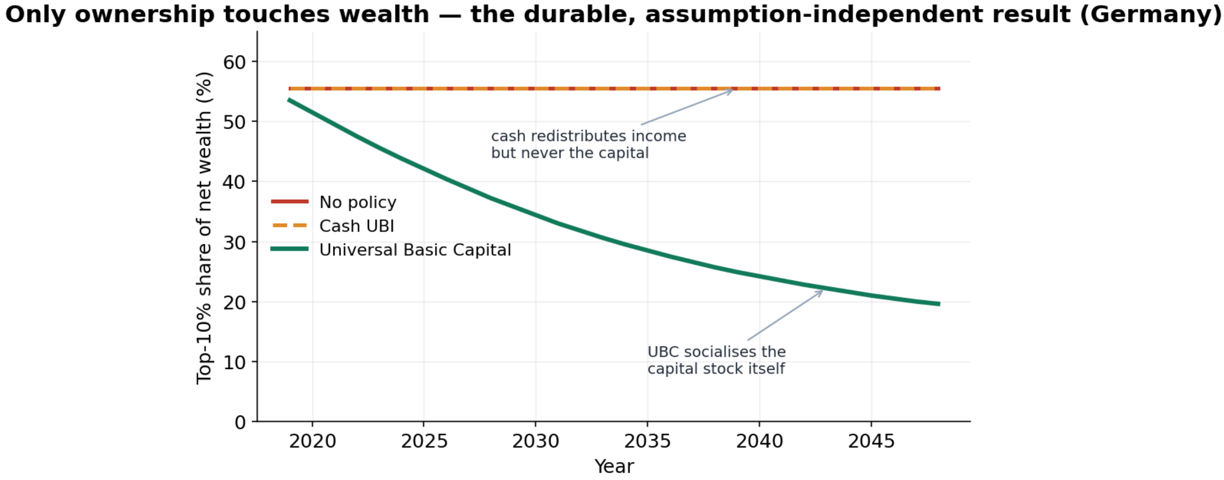

Figure 3 — Only ownership touches wealth. Cash and inaction leave wealth concentration untouched; UBC socialises the capital stock itself.
Source: AGORA distribution module, Germany, τ=40%. Top-10% wealth share anchored to OECD Wealth Distribution Database (live). The result is independent of the model's most uncertain parameter (see Part VI).

The deeper point is that the policy menu is wider than 'cash or nothing'. AGORA treats the question as a menu of mechanisms, each with a real case and a real tension:

The mechanism menu — predistribution versus redistribution

Mechanism	The idea	The tension
Cash UBI	Unconditional income for all.	Redistributes a flow, not the stock; partly recaptured by asset owners through rents.
Universal Basic Capital	Every citizen owns a share of the AI capital; the dividend grows with it.	Fund governance and political capture; how the stake is acquired.
Universal Basic Services / Compute	Distribute the capability (compute, AI access, services) directly.	Can be paternalistic; defining and provisioning the basket.
Data dividends	Pay people for the data that trains AI.	Per-capita value may be small; measurement is hard.
Public-option / commons AI	Build AI as a public utility so surplus is never enclosed.	Who funds the frontier? Same compute-cost concentration.

The cross-cutting distinction is predistribution (change who owns the assets up front) versus redistribution (tax and transfer after the fact). The flow-versus-stock logic points firmly toward predistribution — and UBC is the most direct predistributive instrument.

An honest detour: why we model rather than simply measure

A natural objection is: why not just measure how much AI will shift inequality? Because the market signal is hidden. Across six EU economies, taxes and transfers already erase on average 37% of market inequality (Figure 4) — so the redistribution system absorbs exactly the capital-share movements we want to observe. The sensitivity of inequality to the capital share cannot be cleanly read from recent data; we therefore treat it as an explicit, swappable assumption and test the conclusions across its full plausible range. The headline ownership result (Figure 3) does not depend on it.

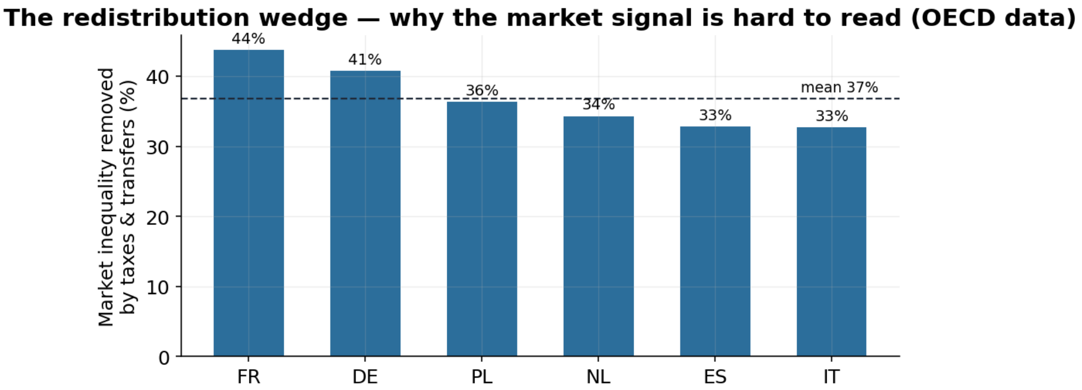

Figure 4 — The redistribution wedge. The share of market inequality already erased by taxes and transfers, by country.

Source: AGORA, OECD Income Distribution Database (market vs disposable Gini). Why the capital-share→inequality link is hard to identify in post-tax data.

PART III

The proposal: Universal Basic Capital

Universal Basic Capital gives every citizen an equal, non-tradable stake in a citizens' sovereign wealth fund that owns a growing share of the economy's capital. Each year, a levy on capital income (set here at τ = 40% for comparability with the cash arm) is converted in kind into the fund — diluting concentrated private ownership rather than collecting a cash tax. The fund holds the assets on behalf of all citizens and pays an equal per-capita dividend. Because the fund's holdings compound, so does the dividend.

AGORA runs cash UBI and UBC at equal cost — the same shock, the same levy intensity, the same government deficit path — so the comparison is clean. The difference is purely flow versus stock, and it reverses over time (Figure 5). Cash helps sooner; UBC compounds and overtakes.

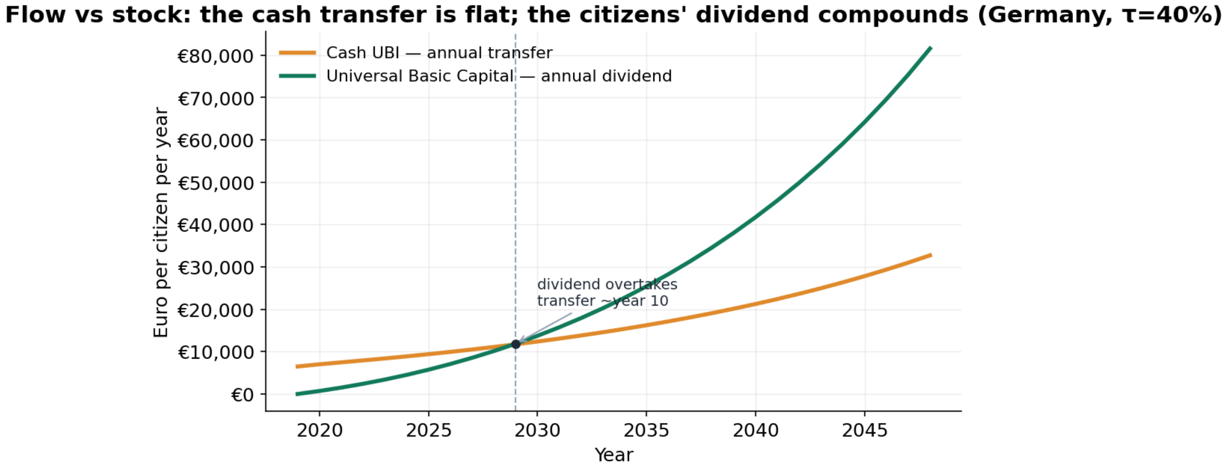

Figure 5 — Flow versus stock. The cash transfer stays flat; the citizens' dividend compounds and overtakes it around year 10.
Source: AGORA UBC experiment, Germany, τ=40%, equal-cost comparison. Per-capita euro values, illustrative scenario.

By the 30-year horizon citizens collectively own about 79% of the capital stock, the per-capita dividend reaches roughly €81,545 per year (against €32,731 for the flat cash transfer), and — the decisive point from Part II — the wealth concentration that cash never touches has been dismantled. Under honest poverty accounting, UBC is the only instrument in the model that drives relative poverty to zero, because the dividend eventually clears the poverty line on its own.

But does giving away ownership choke the investment that drives growth?

This is the central objection to predistribution, and AGORA takes it seriously by letting investment respond to how much capital income owners still retain. The answer (Figure 6) is conditional and instructive. If the fund simply pays everything out, diluting owners does deter private investment and the UBC economy ends smaller (~0.81× the cash economy). But if the fund reinvests part of its returns — which it can, because it owns the capital — it becomes the investor of last resort, and the UBC economy ends larger (~1.11×). The equality verdict holds in every case. C1 — does predistribution choke growth? — is therefore a question of fund design, not a refutation.

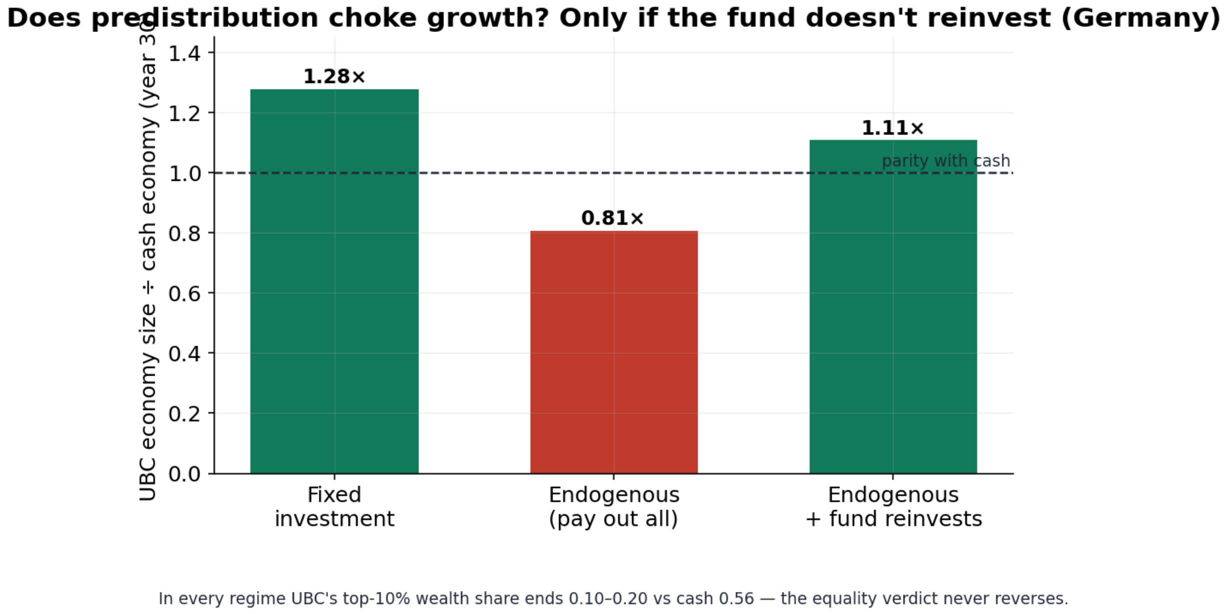

Figure 6 — Predistribution need not cost growth. The UBC economy only shrinks if the fund fails to reinvest; a reinvesting fund ends larger than the cash economy.

Source: AGORA, Germany, investment elasticity 0.75, every scenario consistency-gated. UBC's wealth-equality advantage holds in all three regimes.

The European dimension: a pooled dividend

An AI dividend rests on a common inheritance — collective knowledge, public research, shared data — that is not the property of any one nation. Yet if each member state taxes only its own AI capital and pays only its own citizens, the richer economies throw off bigger dividends and the gap between member states widens. AGORA models the EU as a bloc of separately-gated open economies and compares a national dividend with a pooled one (Figure 7). A purely national scheme entrenches a between-country inequality of ~0.21; pooling the dividend across the Union roughly halves it — under either cash or UBC.

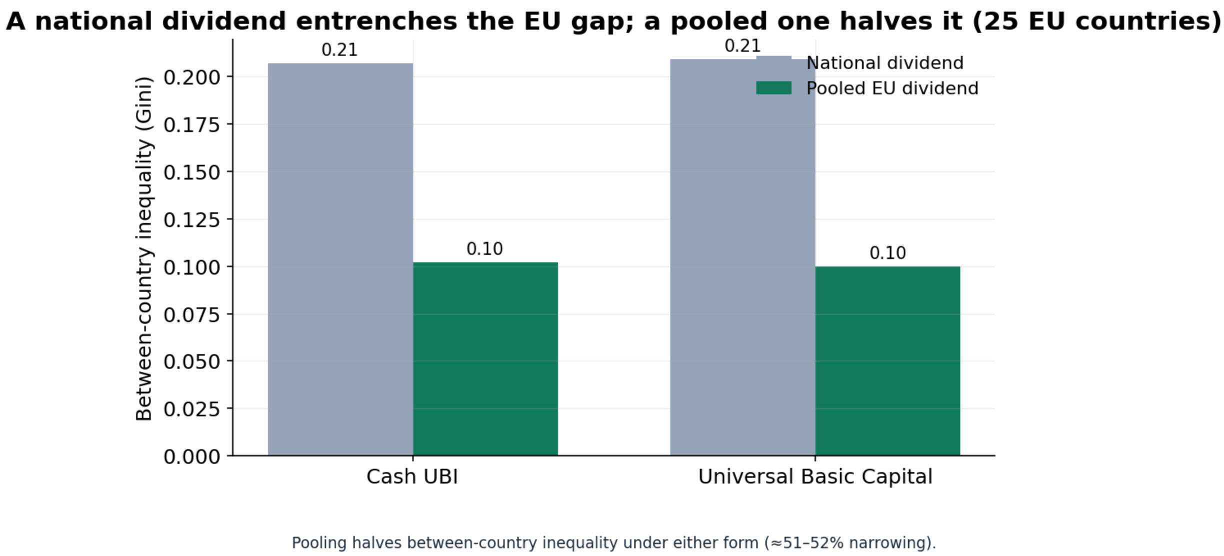

Figure 7 — National entrenches, pooled equalises. A pooled EU dividend cuts between-country inequality roughly in half.

Source: AGORA multi-region module, 25 EU economies, τ=40%, transfers routed through each country's books (gated). 'Universal' is most powerful at Union scale.

PART IV

No single 'best' — but UBC owns the frontier

Honest policy analysis does not crown a single winner; it surfaces trade-offs and leaves the values judgment to citizens and their representatives. AGORA scores every candidate policy on five objectives — growth, equality, poverty, fiscal sustainability and stability — and plots the Pareto frontier: the set of policies where you cannot improve one objective without sacrificing another. Figure 8 shows the growth–equality face of that frontier.

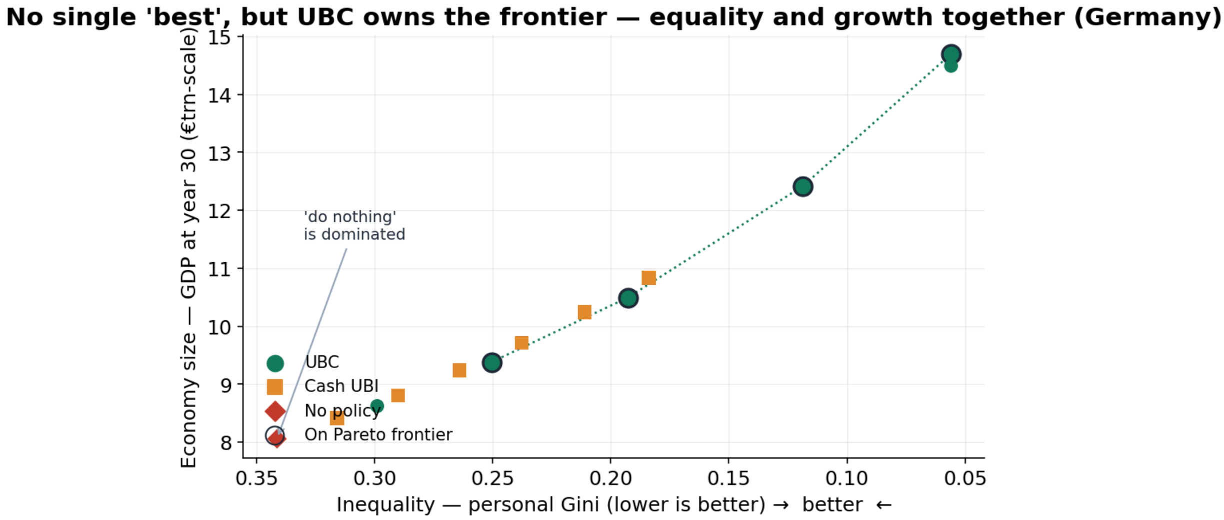

Figure 8 — The policy frontier. UBC policies (circles) dominate both cash UBI (squares) and inaction (diamond); 'do nothing' is dominated outright.

Source: AGORA Phase-4 policy search, Germany; 13 policies, each consistency-gated. Frontier points ringed. Lower Gini and higher GDP are both better.

Two things stand out. First, 'do nothing' is dominated — there exist policies that are better on every objective at once, so inaction is not a neutral default but a positive choice to accept a worse outcome. Second, across the realistic range of levy intensities, the frontier is made of UBC policies: for the same intensity, ownership delivers more equality and more growth than the equivalent cash transfer.

The honest caveat that keeps this from being propaganda

There is a genuine trade-off inside the UBC design. The same reinvestment that makes the economy larger means less of the dividend is paid out as income in the near term — so UBC's income-equality edge over cash is real but conditional on how much is paid out versus reinvested. What is not conditional is the wealth result: ownership moves to citizens in every configuration. The robust claim is about who owns the machine; the income path is a dial society can set.

PART V

Designing the institution

If the instrument is a citizens' fund, the hard problems are institutional, not arithmetic. The model is agnostic about governance; it tells us what the fund must do, not how to keep it honest. Drawing the design implications together:

- Acquisition by dilution, not purchase. The stake is built by converting a capital-income levy in kind into the fund each year, so it requires no upfront public borrowing and grows automatically with the AI capital stock.
- A constitutional dividend-and-reinvestment rule. The split between the per-capita dividend and reinvestment is the single most consequential dial (Part IV). It should be set in law, transparently, and adjusted by rule — not by discretion that invites capture.
- Independence and anti-capture. A citizens' fund concentrates ownership in one institution; its legitimacy depends on political independence, broad mandates that forbid directing capital for patronage, and citizen oversight.
- Equal, non-tradable citizen stakes. Shares are per-capita and inalienable, so the fund cannot re-concentrate through a secondary market — the failure mode of past voucher privatisations.
- Union-scale pooling where the inheritance is shared. Because the AI dividend draws on a common inheritance, a pooled European layer is the instrument that prevents the transition from widening the gap between member states.

> **Transparency as governance — the role of a tool like AGORA**
> A proposal to socialise part of the capital stock will rightly be met with scepticism. The answer is not to ask for trust but to show the mechanism. AGORA is built so that every parameter is inspectable, sourced and swappable; every scenario must pass a consistency gate (no money is created or lost in the accounts); and any baseline is back-tested against real history before scenarios are run on it. A governance debate this consequential should be conducted on a shared, open, auditable model — so that disagreements are about values and assumptions, which are legitimate, rather than about hidden arithmetic.

On sequencing: the levy intensity can start modestly and rise on a pre-announced path; the fund can begin at national scale with a pooled European layer added as coordination allows. None of this requires solving global coordination first — but the European layer is where the Union can lead.

PART VI

What could change this conclusion

Intellectual honesty requires stating what would weaken the case. AGORA is a sandbox; its job is to challenge the thesis, not flatter it. The main caveats:

- These are scenarios, not forecasts. Magnitudes depend on the assumed AI shock (a falling labour share and a capital-investment boom). If automation is mild, the whole problem is smaller — but the ranking of policies is what we rely on, and it is robust across shock sizes.
- The inequality-to-capital-share sensitivity is unidentified in data. As Part II showed, the welfare state hides the market signal. We carry this as an explicit assumption and verified that the cash-versus-UBC verdict holds across its full plausible range; the wealth result does not depend on it at all.
- The 'redistribution can raise output' result is closure-dependent. AGORA's demand-led engine implies that recycling income to high-spending households can lift GDP. This is a modelling choice that should be stress-tested against alternative closures, not trusted blindly.
- Capital mobility and coordination are real constraints. An uncoordinated capital levy can leak across borders; this is an argument for the European (and ultimately broader) scale, and for in-kind dilution over a cash tax that is easier to avoid.
- Fund governance is the binding risk. The economics work; whether a citizens' fund can be kept independent and uncaptured is a political-institutional question this model cannot answer.

The bottom line

Under a wide range of assumptions, the AI transition concentrates ownership, and only an instrument that distributes ownership reaches that dimension. Cash transfers ease the symptom; Universal Basic Capital treats the cause. The numbers here are illustrative; the direction is robust.

Owning the machine

The first industrial revolution made a few people the owners of the machines and turned most people into their employees. The AI revolution will decide the same question again — but faster, and with machines that need far fewer employees. Europe can let AI capital concentrate and then argue forever about how large a cash consolation to pay the people it displaces. Or it can decide, now, that the citizens of Europe are owners of the productive capacity their common inheritance helped create. The tools to model that choice — transparently, honestly, and in full accounting detail — already exist. The choice itself is ours.

APPENDIX

Method, data and validation

The engine. AGORA is a stock-flow-consistent (SFC) model: every euro is someone's asset and someone else's liability, and every scenario must pass a consistency gate (sectoral balances sum to zero; no accounting leaks) before any result is reported. Typical residuals are $\sim 1\times10^{-10}$ of GDP. It integrates a demand-led macro core, a personal income-and-wealth distribution layer, a real input-output (sectoral) layer, an AI-shock driver, a citizens'-fund (UBC) mechanism, and a multi-region layer for the EU bloc.

Calibration & validation. The baseline is calibrated to 2019 data for 26 EU member states (Germany and France in full detail). All thirteen core series are pulled live and current. The model is back-tested against 2010–2019 history: it reproduces realised GDP to roughly 1% mean error. The full automated test suite (192 checks) passes, and every figure in this document is regenerated from the gated engine.

The central AI-shift scenario assumes a sustained capital-investment boom (anchored to observed frontier-compute growth of ~4.2× per year, Epoch AI) and a labour share falling toward 30%, with the fall concentrated in high-automation-exposure sectors. The capital-levy intensity is τ = 40% in the headline comparison; results are shown across τ = 10–60% in the frontier analysis.

Data sources (all integrated through DBnomics unless noted)

Layer / quantity	Source
GDP, consumption, investment, trade (2019)	Eurostat national accounts (nama_10_gdp), live
Labour share (adjusted wage share)	AMECO (ALCD2), live
Household debt (% of GDP)	BIS total credit statistics (WS_TC), live
Top-10% wealth share	OECD Wealth Distribution Database (SH_TOP10), live
Disposable & market income Gini	Eurostat (ilc_di12) + OECD Income Distribution Database, live
Sectoral input-output structure	Eurostat FIGARO symmetric tables (naio_10_cp1700), live
AI / frontier-compute growth	Epoch AI compute trends (grounding the shock)
Government debt, population	Eurostat (gov_10dd_edpt1, demo_gind), live

Reproducibility. Every number in this document corresponds to a specific, gated engine run. The model, its assumptions, and its data connectors are designed to be inspected and challenged. Where a value is observed for a nearby year (the OECD wealth share is a 2021 vintage), this is recorded in the model's provenance.